STACKS, DEFORMATION THEORY: LEMMA-HUGE-DIAGRAM-RINGED-SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1955-2013 of sources/stacks-defos/pinned-original-source.tex; proof span is lines 2015-2155, the complete author proof (139 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 1 of queue_defos.json (pb=5962, stmt 1955-2013, proof 2015-2155).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 08U7: sources/stacks-defos/pinned-original-source.tex, source lines 1935–1943. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08S5: sources/stacks-defos/pinned-original-source.tex, source lines 275–296. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08TN: sources/stacks-defos/pinned-original-source.tex, source lines 2155–2155. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08TL: sources/stacks-defos/pinned-original-source.tex, source lines 2155–2155. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08TM: sources/stacks-defos/pinned-original-source.tex, source lines 2155–2155. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:defos.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Assume given a commutative diagram of morphisms of ringed spaces*

$$
\begin{array}{ccccc}
 & & (X_2, \mathcal{O}_{X_2}) & \xrightarrow[i_2]{} & (X_2', \mathcal{O}_{X_2'}) \\
 & \swarrow^{g} & \downarrow{\scriptstyle f_2} & & \downarrow{\scriptstyle f_2'} \\
 & & (S_2, \mathcal{O}_{S_2}) & \xrightarrow{t_2} & (S_2', \mathcal{O}_{S_2'}) \\
(X_1, \mathcal{O}_{X_1}) & \xrightarrow[i_1]{} & (X_1', \mathcal{O}_{X_1'}) & \swarrow & \\
\downarrow{\scriptstyle f_1} & \swarrow & \downarrow{\scriptstyle f_1'} & & \\
(S_1, \mathcal{O}_{S_1}) & \xrightarrow{t_1} & (S_1', \mathcal{O}_{S_1'}) & &
\end{array}
\tag{0.1.1}
$$

whose horizontal arrows are first order thickenings. Set $\mathcal{G}_j = \mathrm{Ker}(i_j^\sharp)$ *and assume given a* g*-map* $\nu : \mathcal{G}_1 \to \mathcal{G}_2$ *of modules giving rise to the commutative diagram*

$$
\begin{array}{ccccccccc}
 & & & & 0 & \to & \mathcal{G}_2 & \to & \mathcal{O}_{X_2'} & \to & \mathcal{O}_{X_2} & \to & 0 \\
 & & & & & & \uparrow{\scriptstyle c_2} & & \uparrow & & \uparrow & & \\
 & & & & 0 & \to & \mathcal{J}_2 & \to & \mathcal{O}_{S_2'} & \to & \mathcal{O}_{S_2} & \to & 0 \\
 & & & \swarrow & & \swarrow & & \swarrow & & \swarrow & & & \\
0 & \to & \mathcal{G}_1 & \to & \mathcal{O}_{X_1'} & \to & \mathcal{O}_{X_1} & \to & 0 & & & & \\
 & & \uparrow{\scriptstyle c_1} & \swarrow & \uparrow & \swarrow & \uparrow & \swarrow & & & & & \\
0 & \to & \mathcal{J}_1 & \to & \mathcal{O}_{S_1'} & \to & \mathcal{O}_{S_1} & \to & 0 & & & &
\end{array}
\tag{0.1.2}
$$

with front and back solutions to (08U7).

1. *There exist a canonical element in* $\mathrm{Ext}^1_{\mathcal{O}_{X_2}}(Lg^* NL_{X_1/S_1}, \mathcal{G}_2)$ *whose vanishing is a necessary and sufficient condition for the existence of a morphism of ringed spaces* $X_2' \to X_1'$ *fitting into (0.1.1) compatibly with* ν*.*
2. *If there exists a morphism* $X_2' \to X_1'$ *fitting into (0.1.1) compatibly with* ν *the set of all such morphisms is a principal homogeneous space under*

$$\mathrm{Hom}_{\mathcal{O}_{X_1}}(\Omega_{X_1/S_1}, g_*\mathcal{G}_2) = \mathrm{Hom}_{\mathcal{O}_{X_2}}(g^*\Omega_{X_1/S_1}, \mathcal{G}_2) = \mathrm{Ext}^0_{\mathcal{O}_{X_2}}(Lg^* NL_{X_1/S_1}, \mathcal{G}_2).$$

Proof. The naive cotangent complex NL_{X_1/S_1} is defined in Modules, Definition 08TN. The equalities in the last statement of the lemma follow from the fact that g^* is adjoint to g_*, the fact that $H^0(NL_{X_1/S_1}) = \Omega_{X_1/S_1}$ (by construction of the naive cotangent complex) and the fact that Lg^* is the left derived functor of g^*. Thus we will work with the groups $\mathrm{Ext}^k_{\mathcal{O}_{X_2}}(Lg^* NL_{X_1/S_1}, \mathcal{G}_2)$, $k = 0, 1$ in the rest of the proof. We first argue that we can reduce to the case where the underlying topological spaces of all ringed spaces in the lemma is the same.

To do this, observe that $g^{-1} NL_{X_1/S_1}$ is equal to the naive cotangent complex of the homomorphism of sheaves of rings $g^{-1}f_1^{-1}\mathcal{O}_{S_1} \to g^{-1}\mathcal{O}_{X_1}$, see Modules, Lemma 08TM. Moreover, the degree 0 term of NL_{X_1/S_1} is a flat $\mathcal{O}_{X_1}$-module, hence the canonical map

$$Lg^* NL_{X_1/S_1} \longrightarrow g^{-1} NL_{X_1/S_1} \otimes_{g^{-1}\mathcal{O}_{X_1}} \mathcal{O}_{X_2}$$

induces an isomorphism on cohomology sheaves in degrees 0 and -1. Thus we may replace the Ext groups of the lemma with

$$\mathrm{Ext}^k_{g^{-1}\mathcal{O}_{X_1}}(g^{-1} NL_{X_1/S_1}, \mathcal{G}_2) = \mathrm{Ext}^k_{g^{-1}\mathcal{O}_{X_1}}(NL_{g^{-1}\mathcal{O}_{X_1}/g^{-1}f_1^{-1}\mathcal{O}_{S_1}}, \mathcal{G}_2)$$

The set of morphism of ringed spaces $X_2' \to X_1'$ fitting into (0.1.1) compatibly with ν is in one-to-one bijection with the set of homomorphisms of $g^{-1}f_1^{-1}\mathcal{O}_{S_1'}$-algebras $g^{-1}\mathcal{O}_{X_1'} \to \mathcal{O}_{X_2'}$ which are compatible with $f^\sharp$ and ν. In this way we see that we may assume we have a diagram (0.1.2) of sheaves on X and we are looking to find a homomorphism of sheaves of rings $\mathcal{O}_{X_1'} \to \mathcal{O}_{X_2'}$ fitting into it.

In the rest of the proof of the lemma we assume all underlying topological spaces are the same, i.e., we have a diagram (0.1.2) of sheaves on a space X and we are looking for homomorphisms of sheaves of rings $\mathcal{O}_{X_1'} \to \mathcal{O}_{X_2'}$ fitting into it. As ext groups we will use $\mathrm{Ext}^k_{\mathcal{O}_{X_1}}(NL_{\mathcal{O}_{X_1}/\mathcal{O}_{S_1}}, \mathcal{G}_2)$, $k = 0, 1$.

Step 1. Construction of the obstruction class. Consider the sheaf of sets

$$\mathcal{E} = \mathcal{O}_{X_1'} \times_{\mathcal{O}_{X_2}} \mathcal{O}_{X_2'}$$

This comes with a surjective map $\alpha : \mathcal{E} \to \mathcal{O}_{X_1}$ and hence we can use $NL(\alpha)$ instead of $NL_{\mathcal{O}_{X_1}/\mathcal{O}_{S_1}}$, see Modules, Lemma 08TL. Set

$$\mathcal{I}' = \mathrm{Ker}(\mathcal{O}_{S_1'}[\mathcal{E}] \to \mathcal{O}_{X_1}) \quad \text{and} \quad \mathcal{I} = \mathrm{Ker}(\mathcal{O}_{S_1}[\mathcal{E}] \to \mathcal{O}_{X_1})$$

There is a surjection $\mathcal{I}' \to \mathcal{I}$ whose kernel is $\mathcal{J}_1\mathcal{O}_{S_1'}[\mathcal{E}]$. We obtain two homomorphisms of $\mathcal{O}_{S_1'}$-algebras

$$a : \mathcal{O}_{S_1'}[\mathcal{E}] \to \mathcal{O}_{X_1'} \quad \text{and} \quad b : \mathcal{O}_{S_1'}[\mathcal{E}] \to \mathcal{O}_{X_2'}$$

which induce maps $a|_{\mathcal{I}'} : \mathcal{I}' \to \mathcal{G}_1$ and $b|_{\mathcal{I}'} : \mathcal{I}' \to \mathcal{G}_2$. Both a and b annihilate $(\mathcal{I}')^2$. Moreover a and b agree on $\mathcal{J}_1\mathcal{O}_{S_1'}[\mathcal{E}]$ as maps into $\mathcal{G}_2$ because the left hand square of (0.1.2) is commutative. Thus the difference $b|_{\mathcal{I}'} - \nu \circ a|_{\mathcal{I}'}$ induces a well defined $\mathcal{O}_{X_1}$-linear map

$$\xi : \mathcal{I}/\mathcal{I}^2 \longrightarrow \mathcal{G}_2$$

which sends the class of a local section f of $\mathcal{I}$ to $\nu(a(f')) - b(f')$ where f' is a lift of f to a local section of $\mathcal{I}'$. We let $[\xi] \in \mathrm{Ext}^1_{\mathcal{O}_{X_1}}(NL(\alpha), \mathcal{G}_2)$ be the image (see below).

Step 2. Vanishing of $[\xi]$ is necessary. Let us write $\Omega = \Omega_{\mathcal{O}_{S_1}[\mathcal{E}]/\mathcal{O}_{S_1}} \otimes_{\mathcal{O}_{S_1}[\mathcal{E}]} \mathcal{O}_{X_1}$. Observe that $NL(\alpha) = (\mathcal{I}/\mathcal{I}^2 \to \Omega)$ fits into a distinguished triangle

$$\Omega[0] \to NL(\alpha) \to \mathcal{I}/\mathcal{I}^2[1] \to \Omega[1]$$

Thus we see that $[\xi]$ is zero if and only if ξ is a composition $\mathcal{I}/\mathcal{I}^2 \to \Omega \to \mathcal{G}_2$ for some map $\Omega \to \mathcal{G}_2$. Suppose there exists a homomorphisms of sheaves of rings $\varphi : \mathcal{O}_{X_1'} \to \mathcal{O}_{X_2'}$ fitting into (0.1.2). In this case consider the map $\mathcal{O}_{S_1'}[\mathcal{E}] \to \mathcal{G}_2$, $f' \mapsto b(f') - \varphi(a(f'))$. A calculation shows this annihilates $\mathcal{J}_1\mathcal{O}_{S_1'}[\mathcal{E}]$ and induces a derivation $\mathcal{O}_{S_1}[\mathcal{E}] \to \mathcal{G}_2$. The resulting linear map $\Omega \to \mathcal{G}_2$ witnesses the fact that $[\xi] = 0$ in this case.

Step 3. Vanishing of $[\xi]$ is sufficient. Let $\theta : \Omega \to \mathcal{G}_2$ be a $\mathcal{O}_{X_1}$-linear map such that ξ is equal to $\theta \circ (\mathcal{I}/\mathcal{I}^2 \to \Omega)$. Then a calculation shows that

$$b + \theta \circ d : \mathcal{O}_{S_1'}[\mathcal{E}] \to \mathcal{O}_{X_2'}$$

restricted to $\mathcal{I}'$ agrees with $\nu \circ a : \mathcal{I}' \to \mathcal{G}_2$. Since $\mathcal{O}_{X_1'}$ is the pushout of $\mathcal{I}' \to \mathcal{O}_{S_1'}[\mathcal{E}]$ and $\mathcal{I}' \to \mathcal{G}_1$ the maps $b + \theta \circ d$ and a define a map $\mathcal{O}_{X_1'} \to \mathcal{O}_{X_2'}$ fitting into (0.1.2).

Proof of (2) in the special case above. Omitted. Hint: This is exactly the same as the proof of (2) of Lemma 08S5. $\square$